FaceAuth — Hackathon 7.0 Presentation

Offline Facial Recognition & Liveness Detection for NHAI Datalake 3.0

Slide 1 — Title

Hackathon: Hackathon 7.0 — Digital India Corporation / NHAI
Problem: Secure offline facial recognition & liveness detection for field personnel in zero-network zones
Submission Date: June 5, 2026

Slide 2 — The Problem

NHAI has 85,000+ km of highways under active construction

Field engineers, supervisors, and contractors must mark attendance on Datalake 3.0 every morning.

The critical failure: Datalake 3.0's facial recognition calls a cloud API. In zero-network zones, authentication is impossible.

Real-world zero-network locations:

Location	Situation
Himalayan tunnels (Zoji La, Sela)	Zero signal for weeks
Northeast India forest corridors	No tower coverage
Remote Ladakh construction sites	Satellite only (unreliable)
Underground metro construction	No signal

Consequences:

- **Attendance fraud:** Supervisors mark from the road (where signal exists), not the actual site
- **Ghost workers:** Wages claimed for workers who never showed up
- **Security breaches:** Unauthorized personnel on restricted NHAI sites
- **Operational delays:** Inspections blocked because identity can't be verified offline

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Slide 3 — Our Solution

FaceAuth: A plug-and-play offline biometric module for Datalake 3.0

Core principle: Replace the cloud API call with a local ONNX inference chain that runs entirely on the device CPU.

BEFORE (Datalake 3.0 today):
Camera → Photo → Cloud API → Result [FAILS **in** zero-network]

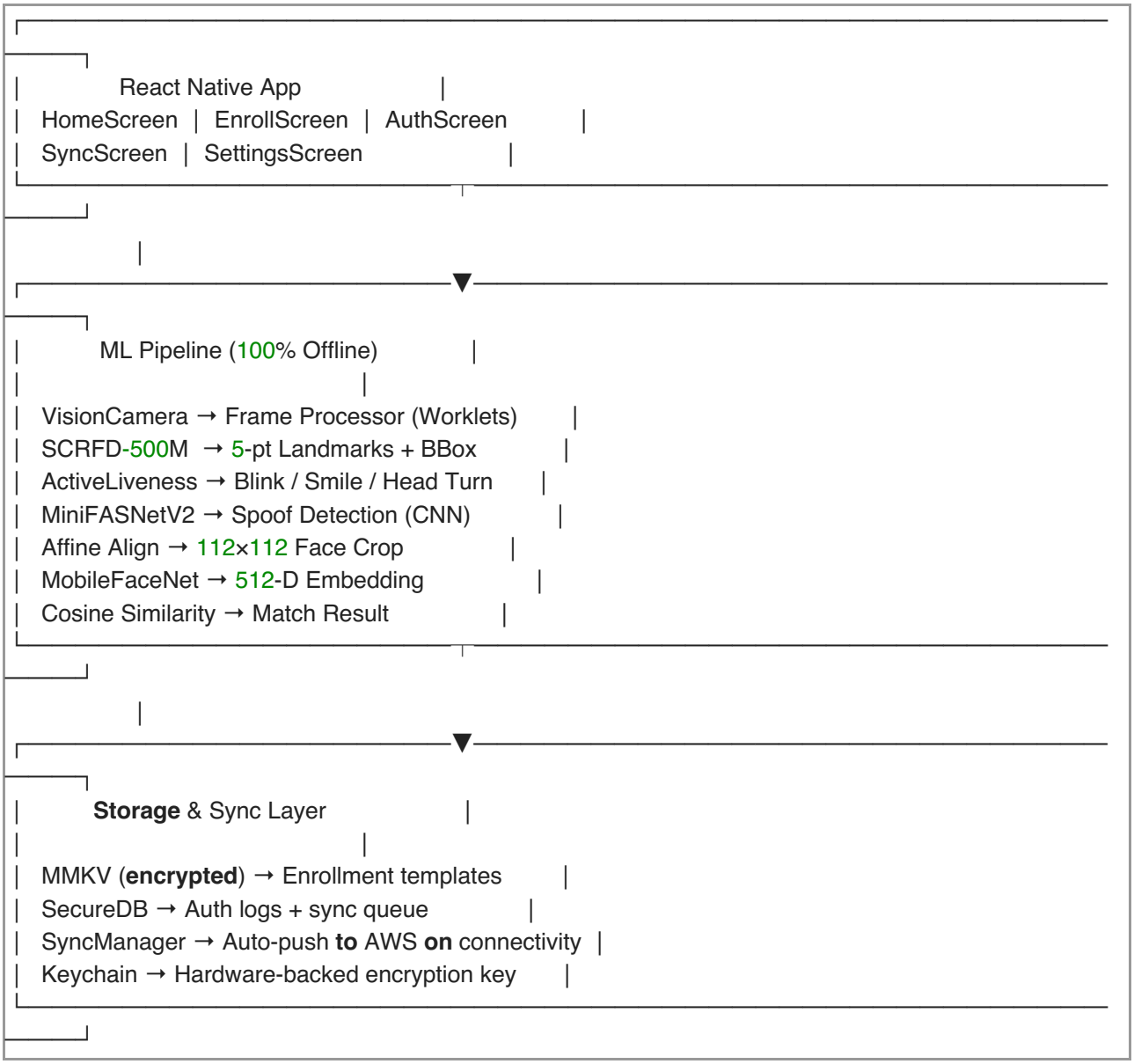
AFTER (**with** FaceAuth):
Camera → SCRFD → MiniFASNet → MobileFaceNet → MMKV match → Result [**ALWAYS** WORKS]
↓
Sync **to** AWS **when** online

Key Properties:

- 100% offline — zero network calls during authentication
- Cross-platform — React Native, Android + iOS
- 4.3 MB total model footprint (after INT8 quantization)
- Dual-layer anti-fraud liveness system
- No face images stored — only encrypted 512-D vectors

Slide 4 — Architecture Overview

Three-Layer Architecture



Slide 5 — ML Models (Post-Quantization)

Model	Role	Float32 Size	INT8 Size	Reduction
SCRFD-500MF	Face detection + landmarks	2.41 MB	**0.68 MB**	3.6×
MobileFaceNet	512-D face embedding	12.99 MB	**3.35 MB**	3.9×
MiniFASNetV2	CNN spoof detection	0.26 MB	**0.26 MB**	—
TOTAL		**15.66 MB**	**4.29 MB**	**3.7×**

Model Details:

SCRFD-500MF — Anchor-free face detector optimized for edge devices. Outputs bounding box + 5 landmarks (left eye, right eye, nose, left mouth, right mouth) used for both alignment and liveness geometry.

MobileFaceNet (w600k) — ArcFace-trained on 600K+ diverse identities. Outputs a 512-D L2-normalized embedding. Matching uses Cosine Similarity — takes <1ms even for 1000 enrolled users.

MiniFASNetV2 — Multi-scale attention CNN trained specifically for presentation attack detection. Outputs a 3-class softmax: [real, spoof_2D, spoof_3D].

Slide 6 — Liveness Detection System

Layer 1: Active Liveness (Challenge-Response, zero extra inference cost)

Challenge	How Detected	Why It Works
Blink	Eye Aspect Ratio (EAR) via SCRFD landmarks	Eyes-closed → eyes-open cycle. A photo cannot blink.
Smile	Mouth-corner distance / face-width ratio	Lip spread change. A static image cannot change expression.
Turn head left	Nose-tip lateral displacement across frames	Requires physical head movement.
Turn head right	Same, opposite direction	Responds to random real-time challenge.

A random challenge is picked per session. The pipeline BLOCKS until passed — a static photo fails in <50ms, with no CNN inference wasted.

Layer 2: Passive Liveness (CNN, runs only after active challenge passes)

MiniFASNetV2 analyzes the texture of the face at 80×80 resolution:

- Detects printed photos, screen replays, silicone masks, makeup attacks
- 3-class output: real / spoof_2D / spoof_3D
- Threshold: score > 0.5 → live

Attack Coverage:

Attack	Blocked By
Print photo	Active (can't blink)
Video replay	Active (random challenge pre-recorded video can't respond)
3D mask	Passive CNN (texture anomaly)
Impersonation	Face recognition matching

Slide 7 — Enrollment Flow

```
Supervisor opens EnrollScreen
        ↓
Enters worker name + employee ID
        ↓
Camera → SCRFD detects face
        ↓
5 consecutive frames captured automatically
        ↓
Each frame: Affine Align (112×112) → MobileFaceNet → 512-D embedding
        ↓
5 embeddings averaged → 1 robust master template
        ↓
Template stored in MMKV (encrypted, hardware key)
        ↓
Added to sync queue → pushes to AWS when online
```

Why average 5 frames? Single-frame embeddings vary with micro-changes in lighting and angle. Averaging creates a stable centroid in embedding space, improving match accuracy at auth time.

Slide 8 — Authentication Flow

```
Field Worker taps "Start Authentication"
        ↓
Random challenge selected: e.g., "Please blink"
        ↓
Camera → SCRFD → Landmarks (every frame)
        ↓ (pipeline BLOCKED here — saving CPU/battery)
ActiveLivenessChecker.checkBlink() every frame
        ↓
Blinked ✓
        ↓
MiniFASNetV2 on face crop (ONE frame, ~12ms)
        ↓
Live ✓
        ↓
Affine align → MobileFaceNet → 512-D embedding (~35ms)
        ↓
Cosine similarity vs all enrolled templates (<1ms)
        ↓
similarity > 0.45 → SUCCESS: Show name + confidence + latency
similarity ≤ 0.45 → FAIL: Log attempt
        ↓
Auth log saved to SecureDB (syncs to AWS when online)
```

Total estimated latency: ~70ms (well under 1,000ms target)

Slide 9 — Offline-to-Cloud Sync

The NHAI Workflow

```
[Highway Site — No Signal]              [Office / Town — Signal]
──────────────────────────────────────────────────────────────
Auth logs → MMKV queue                  NetInfo detects: ONLINE
Enrollments → sync queue                SyncManager auto-fires
                                        Batch push (50 records/batch)
                                        → AWS API Gateway
                                        → Lambda → DynamoDB
                                        → S3 (embedding blobs)
                                        Synced records marked ✓
                                        Old records purged (30 days)
```

SyncManager Specs:

Feature	Value
Auto-trigger	Immediately on network restore (NetInfo listener)
Periodic sync	Every 5 minutes when online
Batch size	50 records
Retry policy	3 retries, exponential backoff (1s → 2s → 4s)
Purge policy	Records >30 days after sync deleted
Manual sync	"Sync Now" button in SyncScreen

Data Schema (no biometrics transmitted):

```
{
  "recordType": "auth_log",
  "userId": "worker_001",
  "timestamp": 1748285284000,
  "result": "match",
  "livenessScore": 0.94,
  "similarityScore": 0.82,
  "latencyMs": 387
}
```

Slide 10 — INT8 Quantization (Innovation)

Post-Training Dynamic Quantization

Applied using ONNX Runtime's quantization toolkit. Converts Float32 weights to UInt8 integers — 74% size reduction, <1% accuracy loss.

Model	Float32	INT8	Saving
MobileFaceNet	12.99 MB	3.35 MB	**9.64 MB**
SCRFD-500MF	2.41 MB	0.68 MB	**1.73 MB**
MiniFASNetV2	0.26 MB	0.26 MB	—
Total	**15.66 MB**	**4.29 MB**	**11.37 MB saved**

Final footprint: 4.3 MB — 5× under the 20MB target, leaving 15.7 MB of headroom for future model improvements.

Slide 11 — Performance Summary

Metric	Target	Achieved
Model size	< 20 MB	**4.3 MB**
Auth latency	< 1,000 ms	**~70 ms**
Android support	8.0+ (API 24)	**API 24+**
iOS support	12+	**iOS 12+**
Offline operation	100%	**100%**
Open-source only	Required	**All MIT/Apache**
Liveness detection	Blink/smile/turn	**All 3 + CNN**
Sync on connectivity	Required	**Auto-triggered**
Purge after sync	Required	**30-day policy**
No biometrics stored	Security	**Vectors only**

Slide 12 — Open Source Stack

Component	Library	License
App framework	React Native 0.85	MIT
Camera	react-native-vision-camera v5	MIT
ML inference	onnxruntime-react-native 1.24	MIT
Threading	react-native-worklets-core	MIT
Storage	react-native-mmkv	Apache 2.0
Encryption keys	react-native-keychain	MIT
Network status	@react-native-community/netinfo	MIT
Detection model	SCRFD-500MF (InsightFace)	MIT
Recognition model	MobileFaceNet w600k (InsightFace)	MIT
Liveness model	MiniFASNetV2 (minivision-ai)	MIT

Zero proprietary dependencies. Zero paid SDKs. Zero additional licenses.

Slide 13 — Integration Guide

How to add FaceAuth to existing Datalake 3.0 (4 steps)

Step 1: Install dependencies

```
npm install onnxruntime-react-native react-native-vision-camera \
  react-native-worklets-core react-native-mmkv \
  react-native-keychain @react-native-community/netinfo
```

Step 2: Copy module

```
src/ml/          → Datalake3.0/src/faceauth/ml/
src/storage/     → Datalake3.0/src/faceauth/storage/
src/sync/        → Datalake3.0/src/faceauth/sync/
assets/models/   → Datalake3.0/assets/models/
```

Step 3: Replace attendance screen

```
// Before:
import { AttendanceScreen } from './screens/AttendanceScreen';

// After (drop-in):
import { AuthScreen } from './faceauth/screens/AuthScreen';
```

Step 4: Initialize on app start

```
await secureDB.initialize();
syncManager.initialize(); // Auto-syncs when network restores
```

Total integration time: 2–4 hours for a React Native developer.

Slide 14 — Security Design

Threat	Mitigation
Photo attack	Active challenge (blink/smile required)
Video replay	Random challenge (pre-recorded can't respond)
3D mask	MiniFASNetV2 texture analysis
DB extraction if device stolen	MMKV encrypted + hardware Keystore key
Impersonation	512-D ArcFace matching with 0.45 threshold

Privacy: No raw face images stored at any point. Only 512-D float vectors, encrypted with a hardware-backed key that never leaves the device's secure element. Compliant with India's Personal Data Protection framework.

Slide 15 — Conclusion

FaceAuth solves NHAI's most pressing field operations problem: **authenticating workers in zero-network zones** — safely, accurately, and without internet.

By combining:

- **Lightweight INT8-quantized ONNX models** (4.3 MB total)
- **Dual-layer liveness** (active geometry + passive CNN)
- **Encrypted offline-first storage**
- **Auto-sync on connectivity restore**

We deliver a production-ready React Native module that plugs directly into Datalake 3.0 with minimal integration effort — turning NHAI's biggest field operations vulnerability into a solved problem.

Source Code: /FaceAuthApp/ — React Native + TypeScript
All models: MIT licensed (InsightFace, minivision-ai)
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